

Sensor, in this case). Define 0 under Channel for the motion sensor and click on Add Actuator.

The two widgets are now added to Cayenne dashboard (Fig. 8).

Next, upload the combined_code.ino sketch to Arduino board for controlling the relay and sensor. However, before doing so, stop the command running on the terminal by typing Q

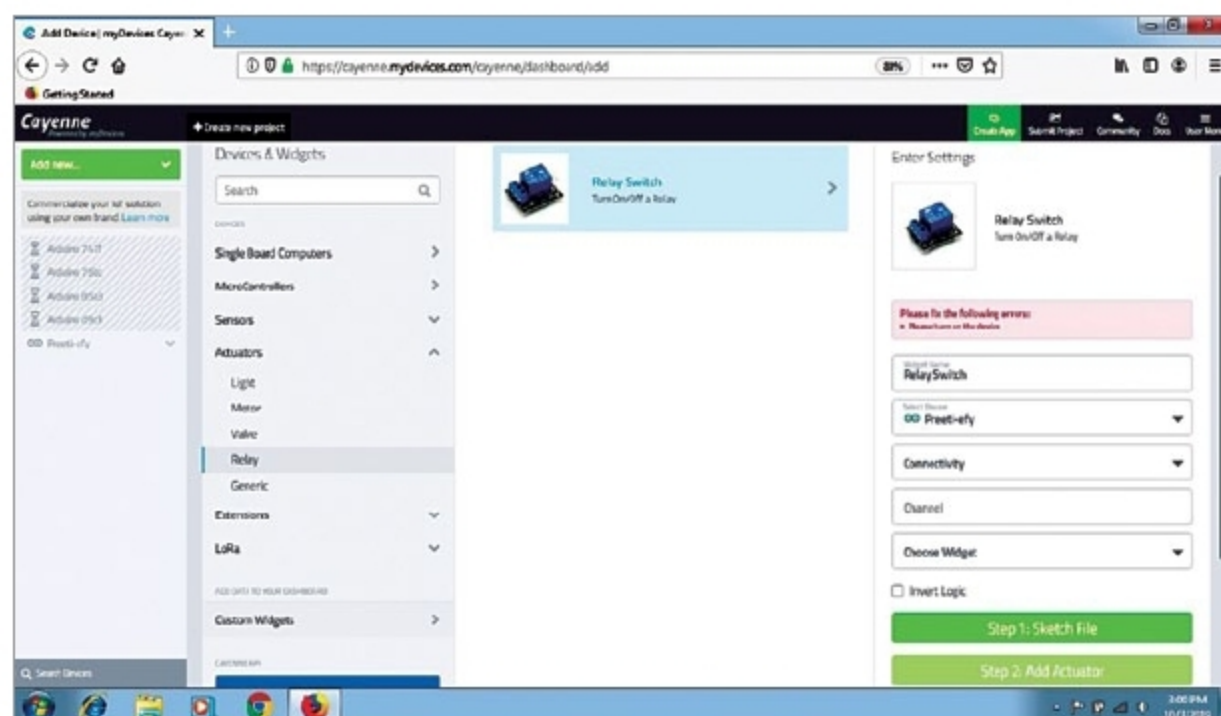


Fig. 7: Relay settings

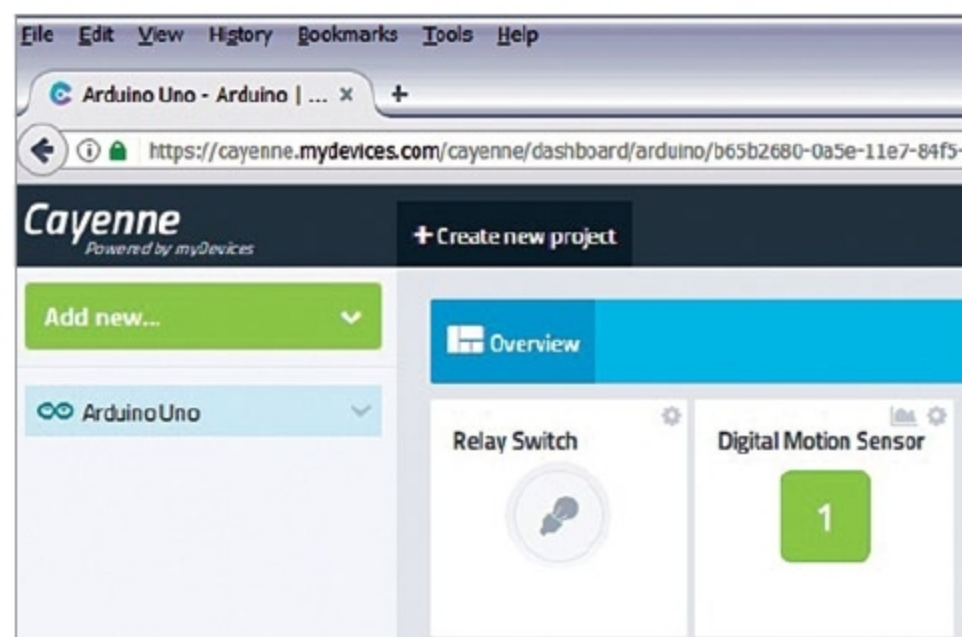


Fig. 8: Widgets on Cayenne dashboard

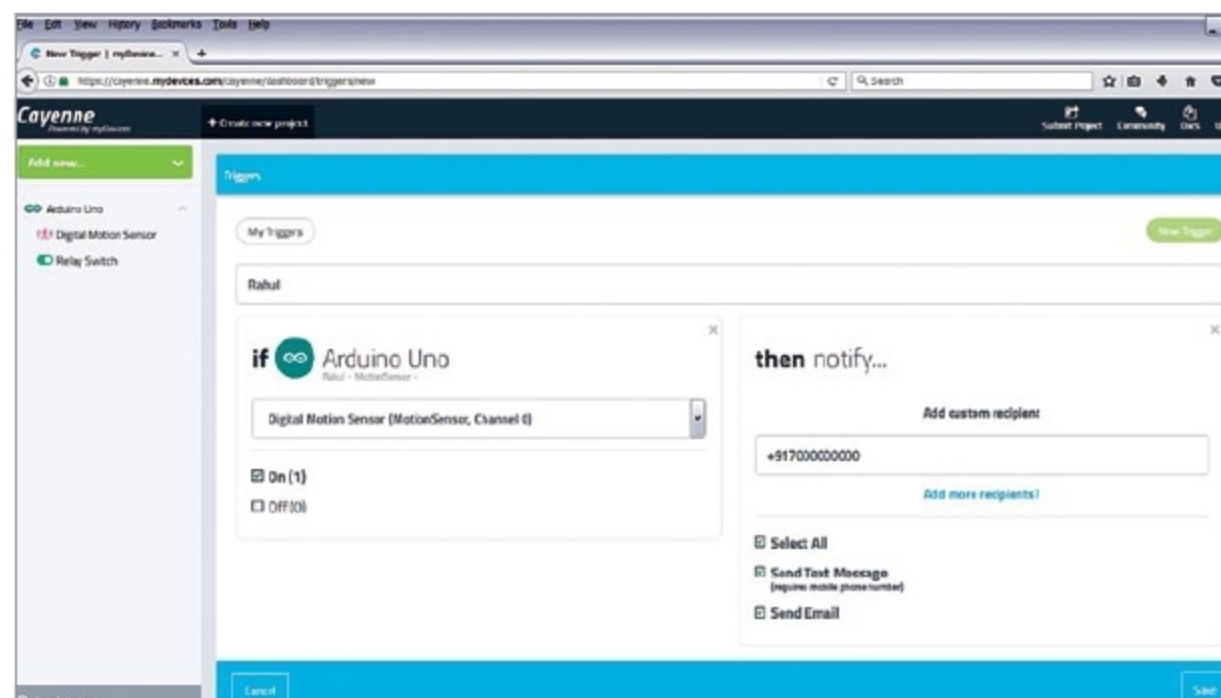


Fig. 9: Trigger settings

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno
- T1 - BC547 npn transistor
- D1 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2 - 220-ohm

Miscellaneous:

- RL1 - 5V single-changeover relay
- BATT.1 - 9V battery
- CON1 - 2-pin connector for load
- CON2 - 2-pin terminal connector for 230V AC mains
- CON3 - 3-pin male Berg connector for PIR sensor
- PIR sensor (HC-SR501)

(All these are available from kitsnspares.com)

etry Transport (MQTT) protocol's username, password and client ID as defined in Serial_USB_Code.ino code. Select proper board and COM port under Tools menu, and compile and upload the combined_code.ino code to Arduino board.

Again run cayenne-ser.bat command to connect Arduino Uno board to Cayenne.

Open Cayenne dashboard to control the relay and sensor via the website. The same can be done using Cayenne app on a smartphone.

There is another feature in Cayenne to activate Trigger function in case of an emergency or to send a security alert. Activating Trigger function will send an SMS alert to the cellphone number or email in case of an emergency. For that, add a cellphone number prefixed with country code, as shown in Fig. 9.

If the PIR sensor detects a human in motion through Arduino Uno, the system will notify you (recipient). You can provide the recipient's cellphone number or email address.

Once you receive an alert via cellphone or email, you can turn on the bulb through the relay switch from Cayenne dashboard.

There are many other features and settings available in Cayenne. It is an interesting and useful platform for alert systems, from surveillance to asset tracking using suitable electronic devices and the Internet. **EFY**



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